



EduPro: Course management and evaluation system

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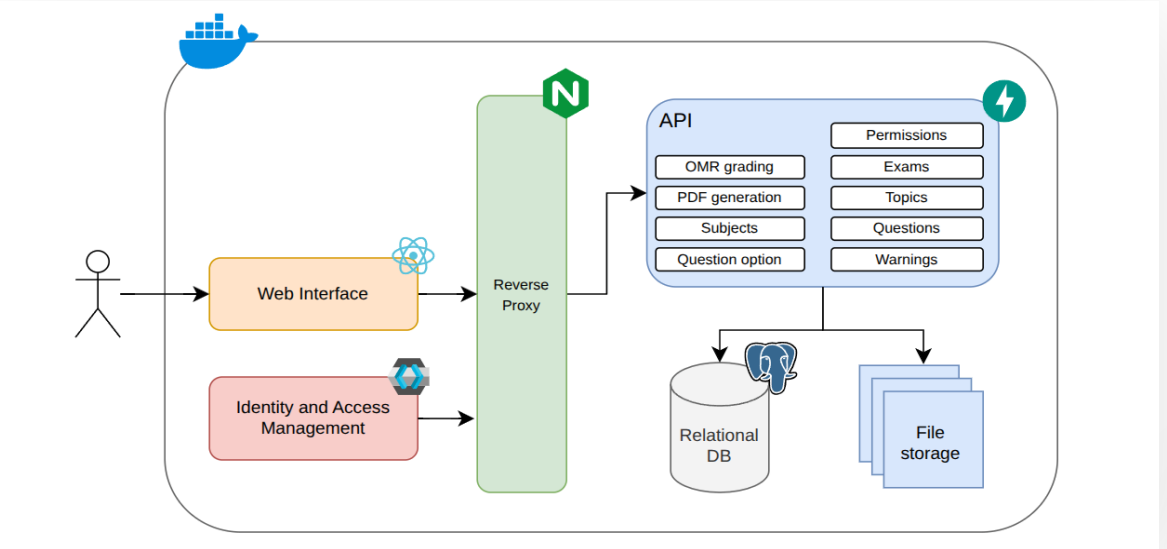


SCAN ME

EduPro is a web application that **automates a course evaluation** pipeline, from the exam generation to the **automatic grading**. It simplifies grading for professors by automating manual tasks through image processing.

Methods

The application follows a layered architecture capable of fast deployment by being containerized.



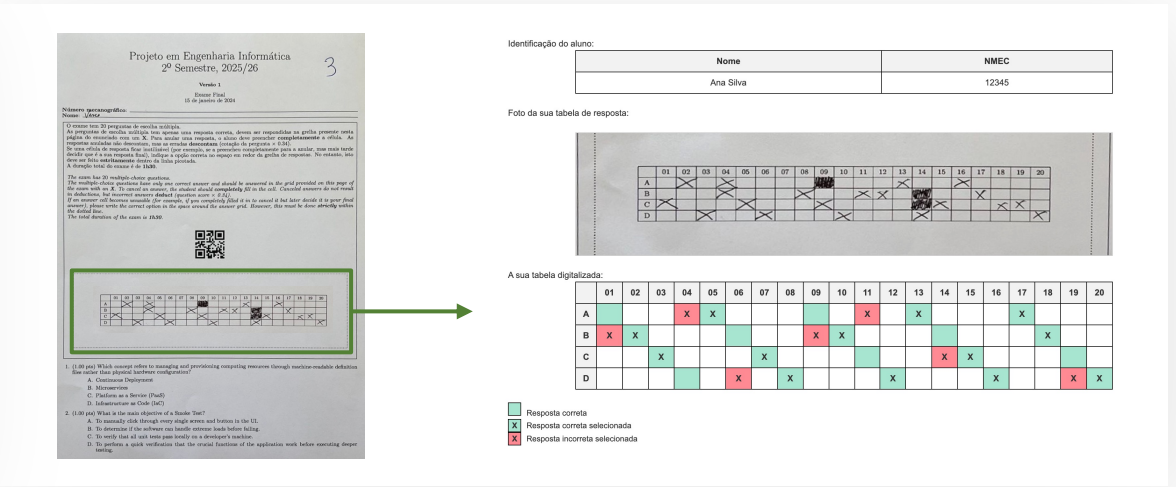
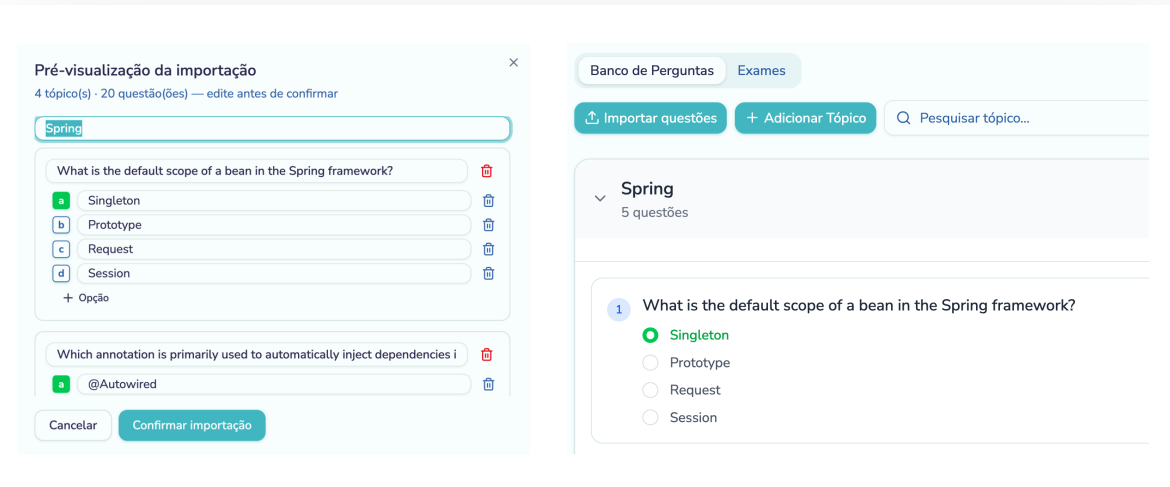
The workflow **shortens the grading moment** by associating a student to an exam, scanning their answers and comparing with the correct solution tables. It is possible to **generate a specific number of versions** of the same exam based on a set of questions.

Results

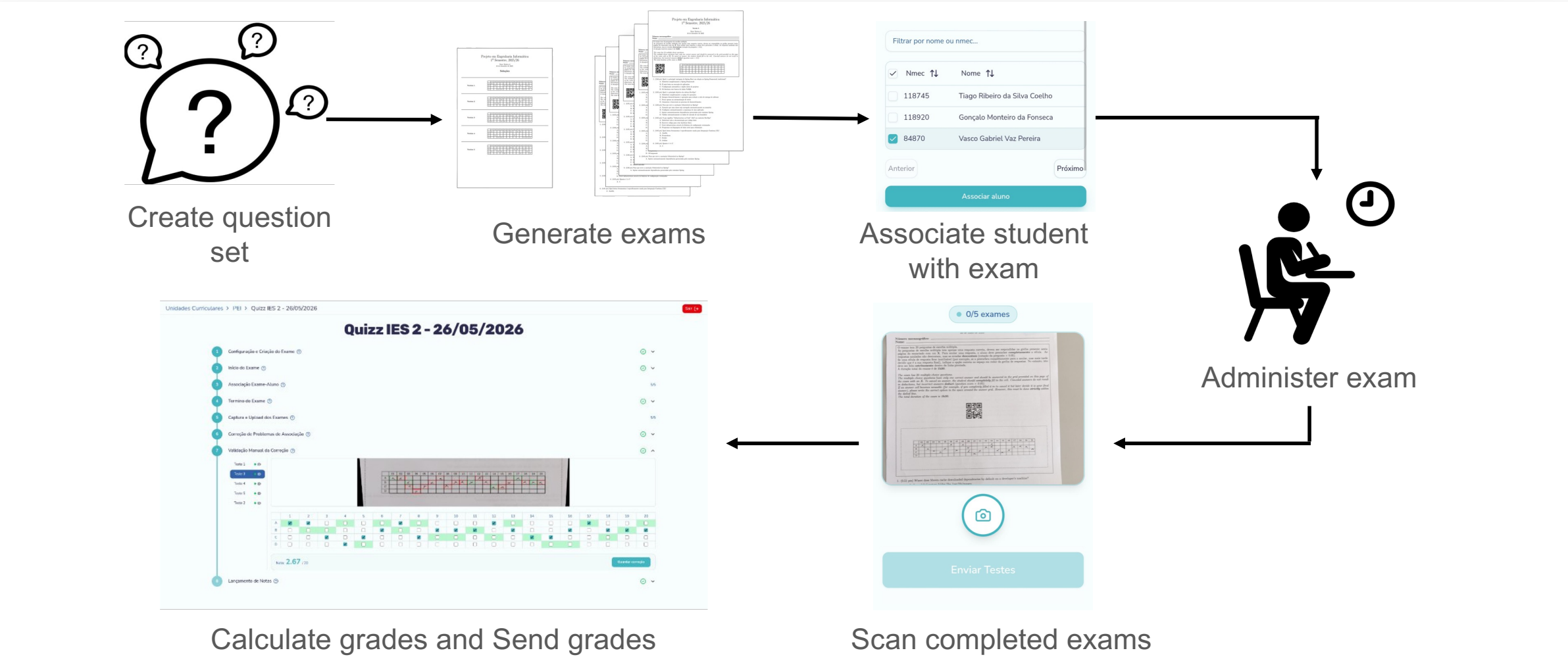
Professors can **fully customise exam generation** by adjusting attributes such as the number of questions, grading weights, and point penalties.

Automatic grading is powered by a custom Convolutional Neural Network (CNN) trained on 6400 samples, where each sample represents a possible exam answer. The model achieved an overall **accuracy of 98.5%** and an F1 score of 97.3%.

Total Detection Errors	Overall Accuracy	Macro Average Precision	Macro Average Recall	Macro Average F1-Score
24	0.9850	0.9662	0.9814	0.9730



System overview



Conclusion: EduPro offers a **complete course management** and **automated grading system**. It simplifies the creation of **unique exams**, replaces manual grading with **instant scanning**, and automatically **delivers grades directly to students**.